

Next-Day Stock Status Prediction in Fresh Retail

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Problem Statement & Challenges

The Fresh Retail Dilemma

Fresh retail operations face a constant battle between **overstocking** (leading to spoilage and massive food waste) and **understocking** (leading to lost revenue, stockouts, and customer churn). Accurate, intraday prediction of out-of-stock (OOS) events is critical for dynamic shelf replenishment.

Core Modeling Challenges

- **Censored Demand (Stockout-Induced Bias):** When an item is out of stock, sales record as zero. Traditional forecasting models misinterpret this as “zero demand”, leading to a negative feedback loop of systematic under-ordering.
- **Severe Class Imbalance:** True stockout hours are rare anomalies (often $< 5\%$ of total operational hours), making it extremely difficult for standard ML loss functions to learn the minority class without collapsing.
- **High-Velocity Intraday Dynamics:** Unlike durable goods, fresh produce sales exhibit extreme intraday volatility driven by time-of-day foot traffic, flash sales, and perishable shelf-life windows.
- **Hierarchical Spatial-Temporal Variance:** Demand shocks are not uniform; they vary heavily across different city zones, specific store demographics, and individual SKU categories.

Literature Review & Related Work

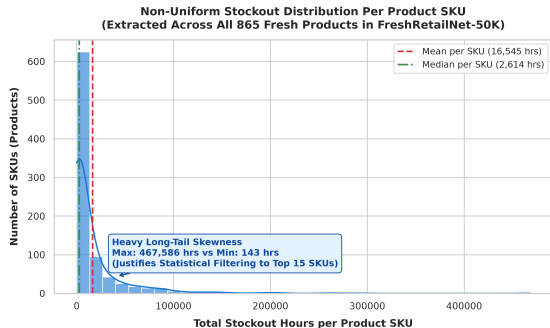
General Stockout Prediction Research

- **Classical ML Ensembles** (Andaur et al., 2021): Showcased Random Forest and voting ensembles as robust benchmarks for retail stockout prediction on tabular data.
- **Deep Learning Architectures** (Giaconia & Chamas, 2023): Utilized massive CNN, Self-Attention, and Temporal Convolutional Networks (TCN) (> 500k parameters) for intraday stockout probability.
- **Large-Scale Imbalance Handling** (Liu et al., 2025): Benchmarked XGBoost and SMOTE strategies, proving that near-term demand indicators dominate long-term forecasts for stockout events.

Research Utilizing the FreshRetail Dataset

- **Hierarchical Forecasting**: *"A Coupled Hierarchical Architecture for Multi-Granularity Demand Forecasting"* uses this exact dataset to tackle the complex spatial/product hierarchy (City → Store → SKU).
- **Censored Demand Bias**: *"When Less Data Is Better: Eliminating Stockout-Induced Bias in Choice Estimation"* explicitly models how stockouts artificially deflate true demand signals in FreshRetail environments.

Dataset Overview & Processing Strategy



FreshRetailNet-50K Stockout Distribution across 865 SKUs

FreshRetailNet-50K Scale & Scope

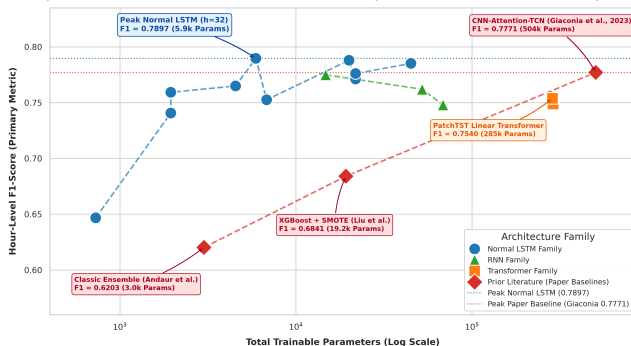
- **Volume:** 50,000 store-product pairs across 898 stores.
- **Timeframe:** 90-day series of granular hourly data (75d Train / 15d Val).
- **Feature Types:** Includes sales history, stockout status, promotions, weather metrics, calendar signals, and momentum streaks.

Why We Selected This Dataset

- **Pioneering Benchmark:** Designed for **censored demand estimation** in fresh retail (where out-of-stock items hide true demand).
- **Targeted Selection:** Isolated the **Top 15 SKUs** using a custom ranking formula to focus on high-impact goods and eliminate noise.

Model Family Comparison: Simple LSTMs Beat Heavy Transformers

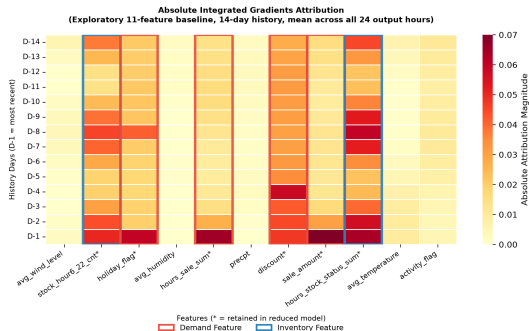
Hourly Stockout Prediction: Parameter-Efficient Normal LSTM Outperforms Transformers, RNNs & Prior Paper Baselines



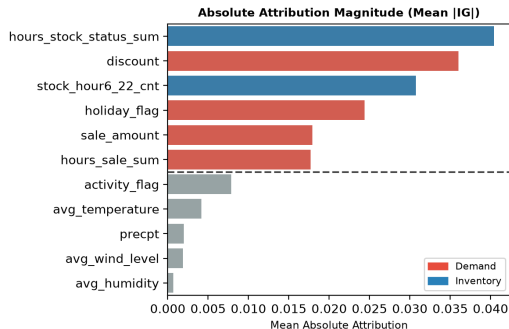
Key Empirical Findings

- **Parameter Efficiency:** Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization:** PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win:** Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability & Baseline Limitations



Attribution heatmap: 11-feature baseline (D-1 = most recent)



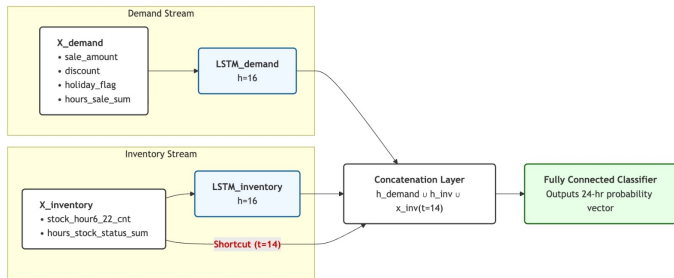
Feature importance: top 6 (highlighted) outrank remaining 5

Motivation for Dual-Stream Architecture

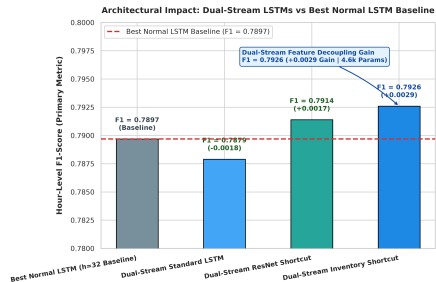
- **F1 Plateau:** Baseline LSTM peaks at ~ 0.789 F1; simply scaling parameters yields no gain.
- **Interference:** Sharing a single hidden state h_t creates potential for demand/inventory interference.
- **IG Analysis:** Absolute attribution reveals distinct temporal behaviors for demand vs. inventory.

These findings motivate processing both streams independently before a task-specific fusion layer.

Our Fix: Dual-Stream Inventory Shortcut LSTM



Dual-Stream: Demand LSTM + Inventory LSTM + Day-14 Shortcut



Architectural Impact: Dual Stream LSTM

Key Results

- **Inventory Shortcut:** F1 **0.7926**, MAE **3.97 hrs** at just 4,600 params.
- Decoupled streams consistently beat all baseline LSTM sizes.

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

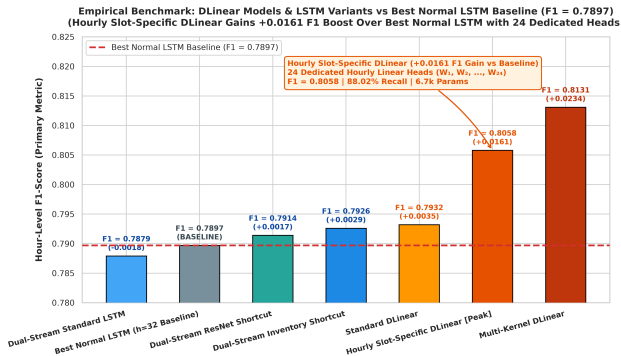
$$X_{\text{trend}} = \text{AvgPool}(X), \quad X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$Y_{\text{pred}} = W_{\text{trend}} X_{\text{trend}} + W_{\text{seasonal}} X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters:** 6,768 parameters.
- **Validation F1-Score: 0.7932**
 - Outperforms Normal LSTMs (**0.7897**)
 - Outperforms Enhanced LSTMs (**0.7926**)
 - Outperforms Transformers (**0.7540**)
- **Stockout Recall: 87.44%.**

Architectural Innovation 2: Linear Changes in DLinear



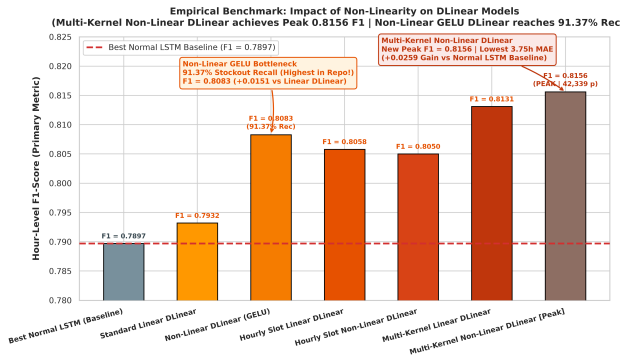
Hourly Slot-Specific Linear DLinear

- Uses **24 dedicated linear heads** ($W_1, \dots, W_{24}$) for each operational hour.
- Resolves inter-hour gradient conflict between morning and evening rush hours.
- Boosts F1 to **0.8058** (+0.0161 over LSTM).

Multi-Kernel Linear DLinear

- Multi-scale moving average trend pooling ($k \in \{3, 5, 7\}$).
- Reaches **0.8131 F1-Score** with pure linear decomposition (13.5k params).

Architectural Innovation 3: Introducing Non-Linearity into DLinear



Non-Linear Activation Bottleneck

Adds GELU activation bottlenecks between decomposition and output projections:

$$\hat{Y} = W_{\text{trend},2} \text{GELU}(W_{\text{trend},1} X_{\text{trend}}) + \text{Seasonal}$$

Breakthrough Results

- **Non-Linear GELU DLinear:** Reaches **91.37% Stockout Recall** (Highest in Repo!) and **0.8083 F1**.
- **Multi-Kernel Non-Linear DLinear:** Reaches **0.8156 F1-Score** (New Peak F1 & lowest 3.75h MAE).

Comparative Analysis

Rank	Architecture Name	Category	Params	F1-Score	Recall
1	Multi-Kernel Non-Linear DLinear	Non-Linear DLinear	42,339	0.8156	87.13%
2	Multi-Kernel Linear DLinear	Linear (Multi-Scale)	13,539	0.8131	85.22%
3	Non-Linear GELU DLinear	Non-Linear DLinear	21,168	0.8083	91.37%
4	Hourly Slot-Specific DLinear	Linear (24 Heads)	6,744	0.8058	88.02%
5	Standard Linear DLinear	Linear (Decomp)	6,768	0.7932	87.44%
6	Dual-Stream Inventory Shortcut	LSTM Family	4,600	0.7926	80.79%
7	Best Normal LSTM (h=32)	Normal LSTM	5,912	0.7897	81.29%
8	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%
9	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%
10	CNN-Attention-TCN (Giaconia et al., 2023)	Deep CNN-TCN	504,088	0.7771	85.45%
11	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%
12	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%
13	XGBoost + SMOTE (Liu et al., 2025)	GBDT / Resample	19,200	0.6841	–
14	Classic Ensemble (Andaur et al., 2021)	ML Ensemble	3,000	0.6203	–

Conclusion and Future Scope

Key Conclusions

- **Lightweight Linear Decompositions Win:** DLinear variants (13k–42k params) outperform heavy Transformers ($> 280k$ params) and deep CNN-TCN models ($> 500k$ params).
- **Dedicated 24 Slot Heads:** Resolves inter-hour gradient conflicts across operating store hours (+0.0161 F1 boost).
- **GELU Non-Linearity Bottleneck:** Captures non-linear inventory step-functions, achieving peak **0.8156 F1** and **91.37% Stockout Recall**.

Future Scope

- **Intraday Streaming Data:** Incorporate real-time point-of-sale transaction streams for dynamic intra-day stockout threshold adjustments.
- **Spatial-Store Hierarchy:** Scale hourly slot projections across regional warehouse clusters for multi-echelon inventory replenishment.
- **Censored Demand Modeling:** Integrate explicit choice estimation to unbiased true demand during zero-stock hours.